

Day 03 Agenda

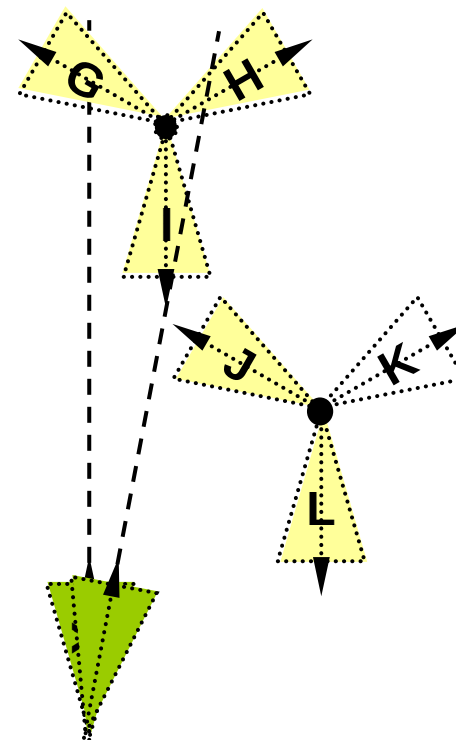
- **RNO:FAS/FOX**
- **Exercises**

FAS

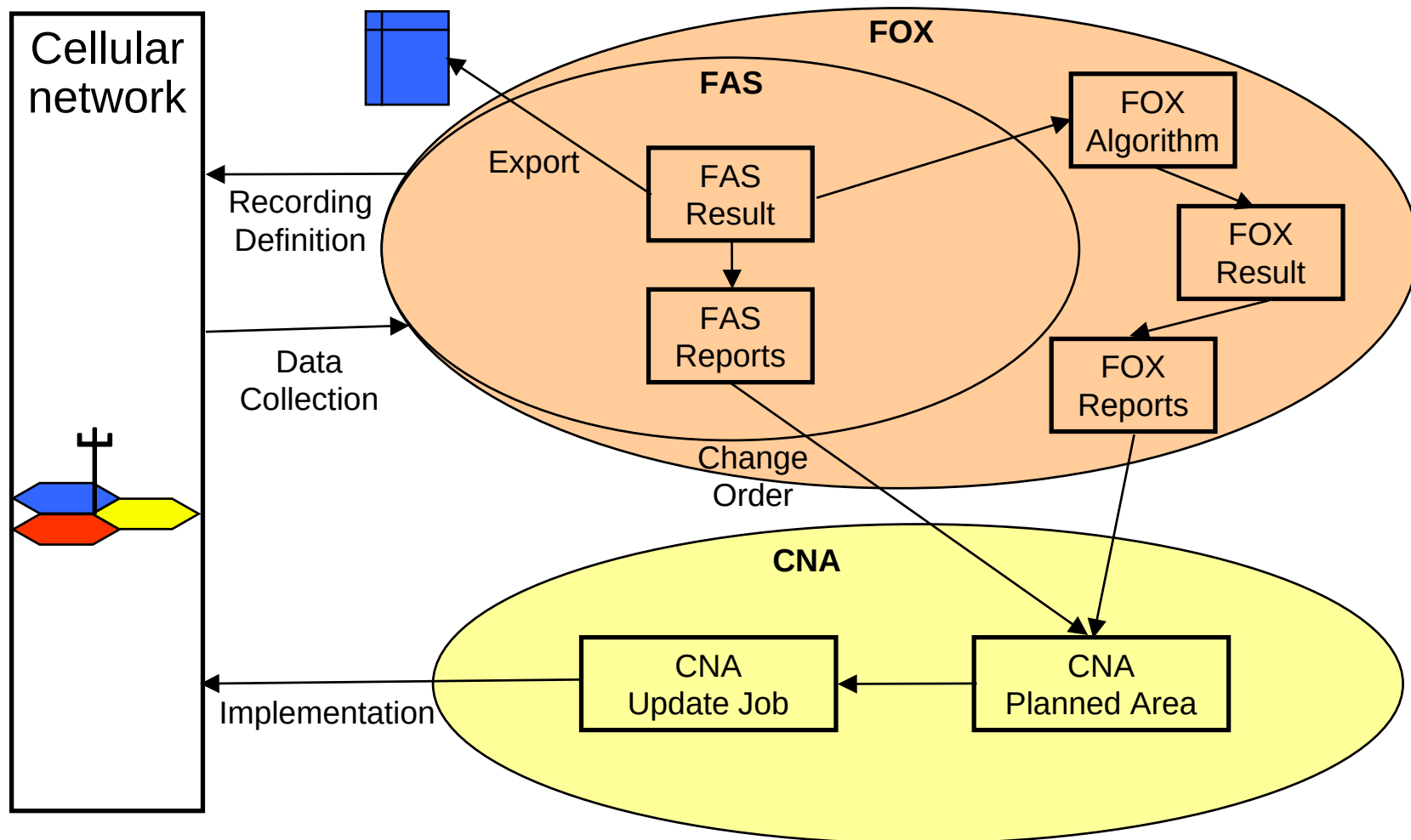
- **FAS is a tool intended to support frequency optimization in order to minimize interference in the radio network.**
- The interference level in GSM networks has to be kept to a minimum in order to use the frequency spectrum in an efficient way, and to increase the capacity of the network. The speech quality and the amount of dropped calls are directly affected by the interference level.

FAS/FOX - Examples of usage

- Identify bad BCCH/TCH frequency
- Detect external interference
- Antenna in wrong direction
- Areas with no dominant server
- Cell coverage other than expected
- Swapped sectors
- Cells built in location different from what was planned
- Location, frequencies and neighbours for new cell and sites



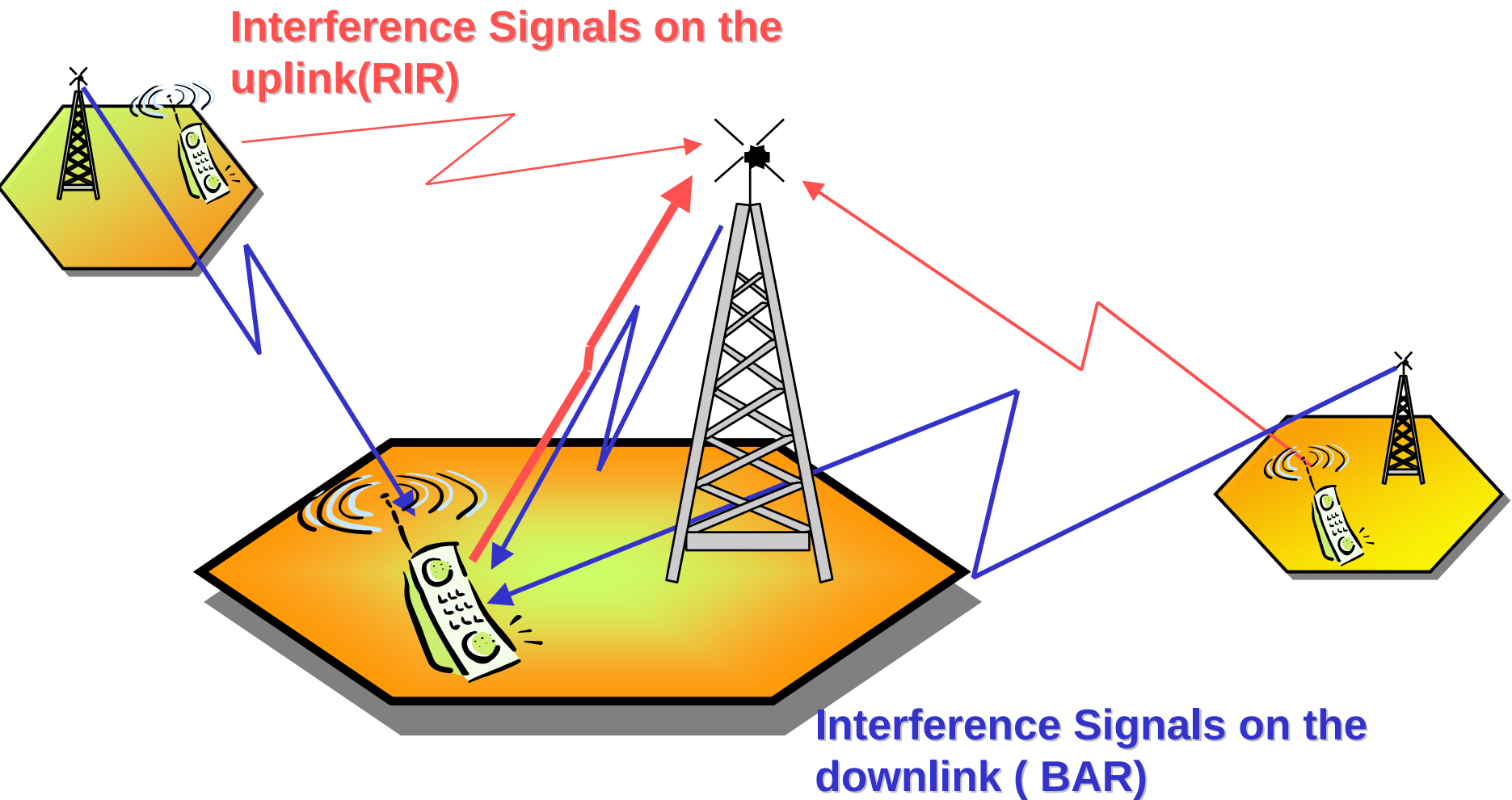
Relation Between FAS and **FOX !!!!**



FAS performs the following activities:

- **Uplink recording**
- **Downlink recording**
- **Report Presentation**
- **Export of Results**

FAS/FOX - Frequency allocation support/ eXpert



RNO – FAS New Recording

Recording Name:

☐ Create New ICDM
☒ Create FAS Result
☒ Estimate PIT

ICDM:

Percentile: %
Relative SS Threshold C/I: dB
Relative SS Threshold C/A: dB

BA List Change Interval (minutes):
No. of Test Freqs. to Add at Each Interval:

Time

Start Date:

Repeat: **Start Date Only** **Times**

Days: **Workingdays** ☐ Sun ☐ Mon ☐ Tue ☐ Wed ☐ Thu ☐ Fri ☐ Sat

Hours: - - - -

Cell Set

Cell Filter: **GSM900**

☐ East
☒ M01B00

- ☒ S1000C0 (GSM900)
- ☒ S1001C0 (GSM900)
- ☒ S1002C0 (GSM900)
- ☒ S1003C0 (GSM900)
- ☒ S1004C0 (GSM900)
- ☒ S1005C0 (GSM900)
- ☒ S1006C0 (GSM900)

Uplink Frequencies

Full Set

Comments

Automatic Export

☐ Recording Results

FAS - Important Recording Parameters

RNO – FAS New Recording

Recording Name: OldTown

☐ Create New ICDM

☒ Create FAS Result

☒ Estimate PIT

ICDM:

ICDM Settings...

PIT Estimates Settings...

Select ICDM...

Percentile: 95 %

Relative SS Threshold C/I: -9 dB

Relative SS Threshold C/A: 9 dB

BA List Change Interval (minutes): 10

No. of Test Freqs. to Add at Each Interval: 5

Create New ICDM/Create FAS Result

- The user can specify whether to record on:
 - Uplink to get a FAS-result,
 - on Downlink to get an ICDM-result
 - or on both Uplink and Downlink.

Relative Signal Strength Threshold C/I

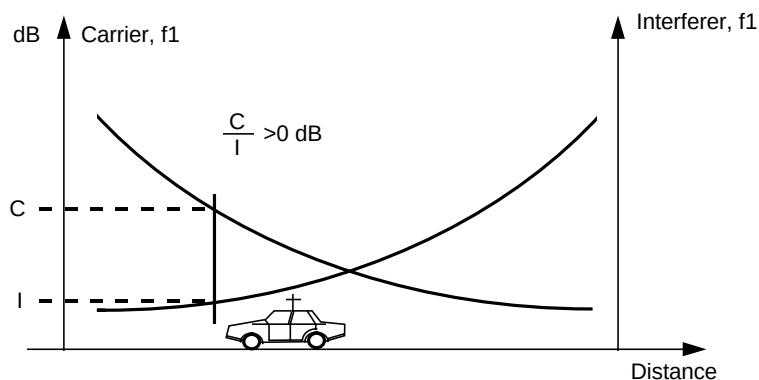
- determines the relative C/I threshold for a neighbouring cell when it is considered to cause **co-channel interference** in the measuring cell.
- $SS_{neigh,BSIC} > SS_{serv} + \text{Relative SS Thresh C/I}$
f f
- **This parameter must be defined for downlink recordings.**
- The values range from -63 to +63 dB.

Relative Signal Strength Threshold C/A

- determines the relative C/A threshold for when a neighbour cell is considered to cause **adjacent-channel** interference in the measuring cell.
- $SS_{\text{neigh,BSIC } f \pm 200\text{kHz}} > SS_{\text{serv } f} + \text{Relative SS Thresh C/A}$
- **This parameter must be defined for downlink recordings.**
- The values range from -63 to +63 dB.

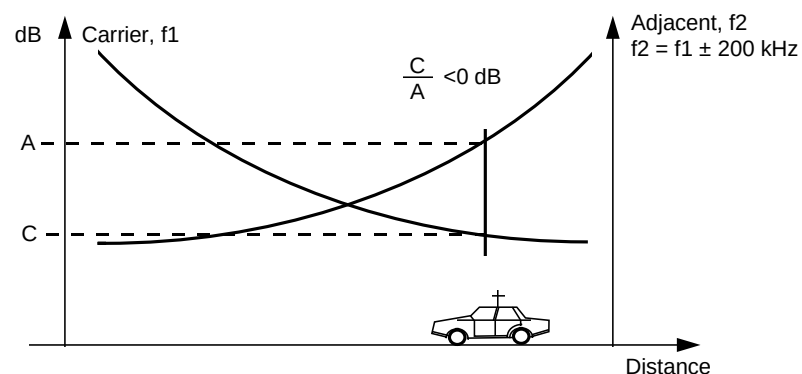
Interference

Co-channel



- $C/I > 9 \text{ dB}$ (GSM specification)
- $C/I > 12 \text{ dB}$ (Ericsson recommended)
- $C/I > 9 \text{ dB}$ (If Fq. hopping is used)

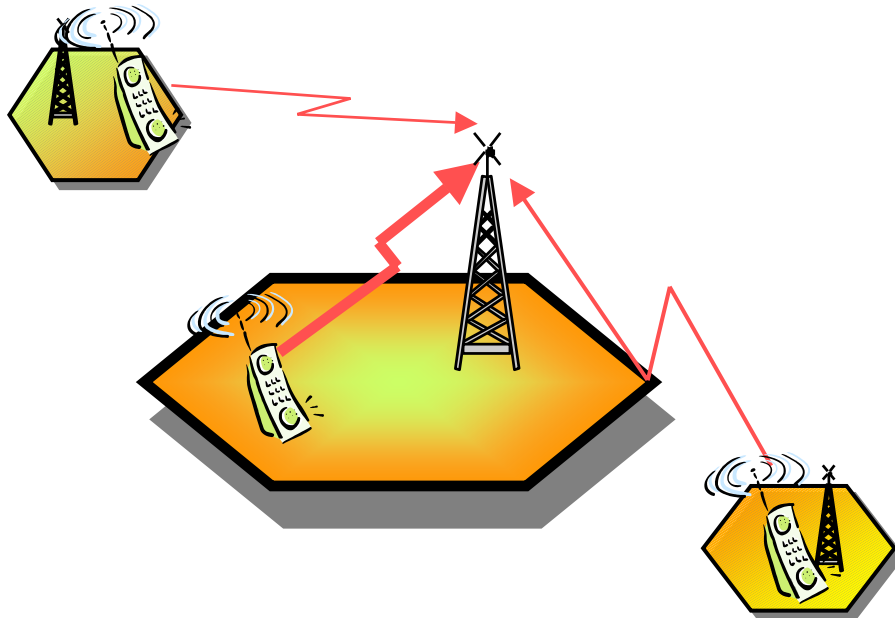
Adjacent channel



- $C/A > -9 \text{ dB}$ (GSM specification)
- $C/A > 3 \text{ dB}$ (Ericsson recommended)

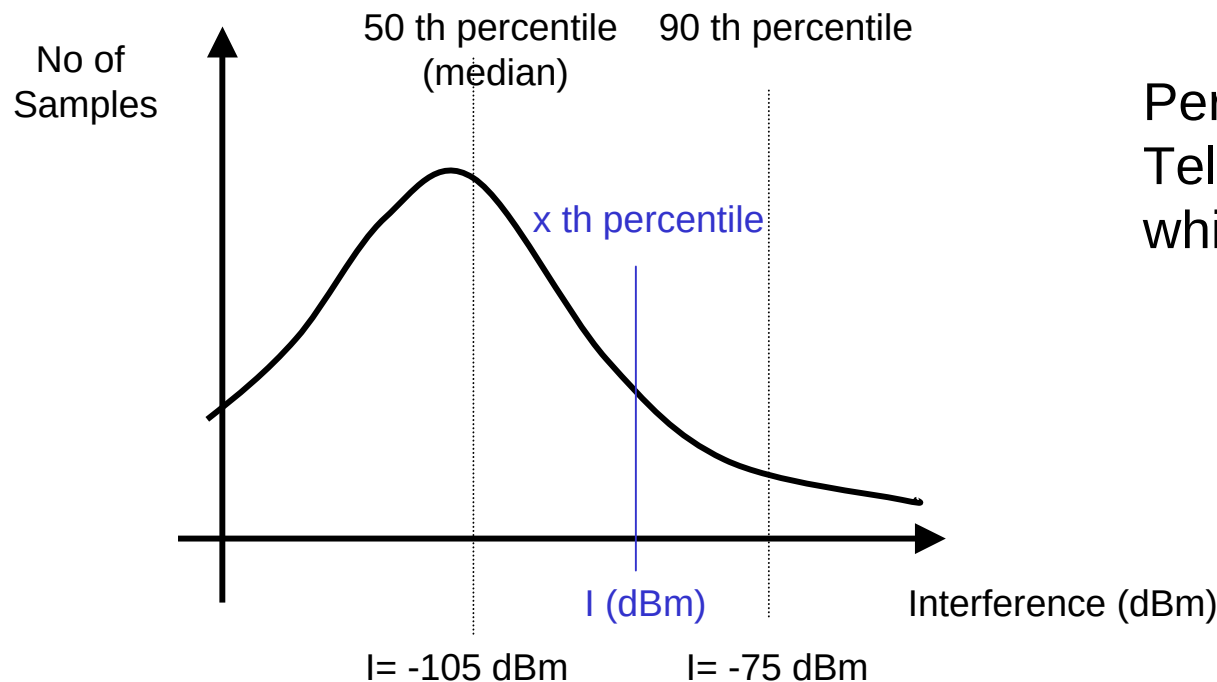
FAS/FOX

How to see uplink Interference ?



- TRX measures the uplink interference repeatedly on configured and test frequencies
- Samples every 15 sec
- Use RIR function in BSC

Distribution of uplink interference



Percentile:
Tells the value $I \text{ dBm}$ below which $x\%$ of the samples fall

Percentile

- A percentile is a set of measured interference levels that has a value below a defined percentile value.
- In a FAS-recording the percentile value is specified when defining a recording.
- The Network Element calculates the highest measured interference level in this percentile.

Recording Percentile

- The percentile value from the interference distribution that is reported for each measured frequency. Given as a parameter by the user.

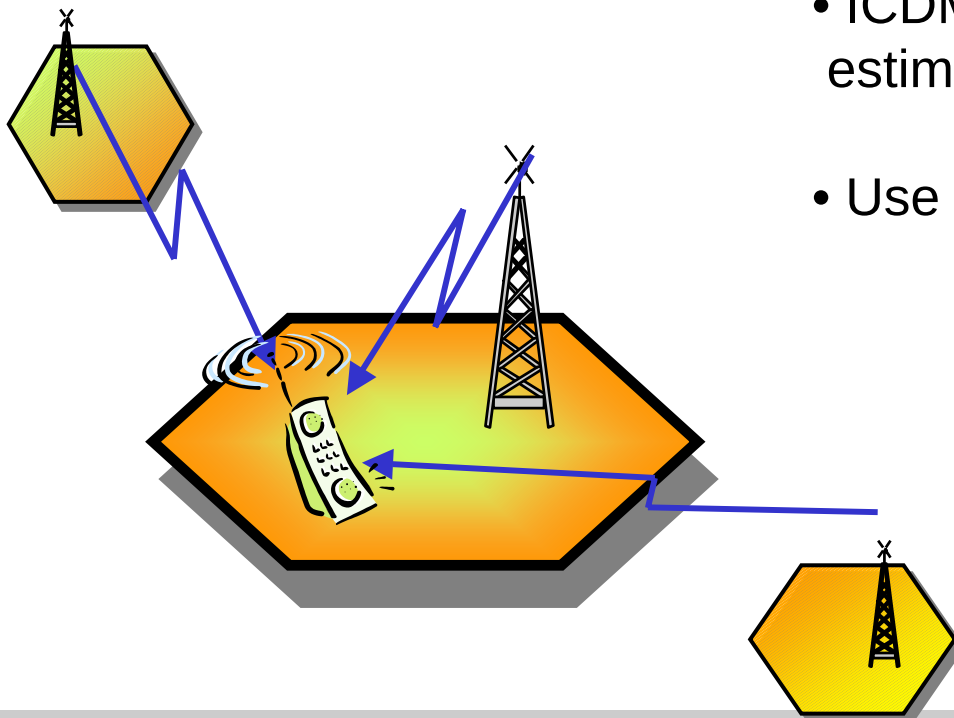
Creating an Uplink Recording

- Create FAS Result checkbox is selected.
- As default the Estimate PIT checkbox is also selected which requires an ICDM and the PIT Estimate settings as input.
- **Deselect the Estimate PIT checkbox to have the recording only defined on the uplink.**

FAS/FOX

How to see downlink Interference ?

- Measurement Reports from MS on configured and test frequencies are collected
- ICDM & PIT are computed to give estimation of downlink interference
- Use BAR function in BSC



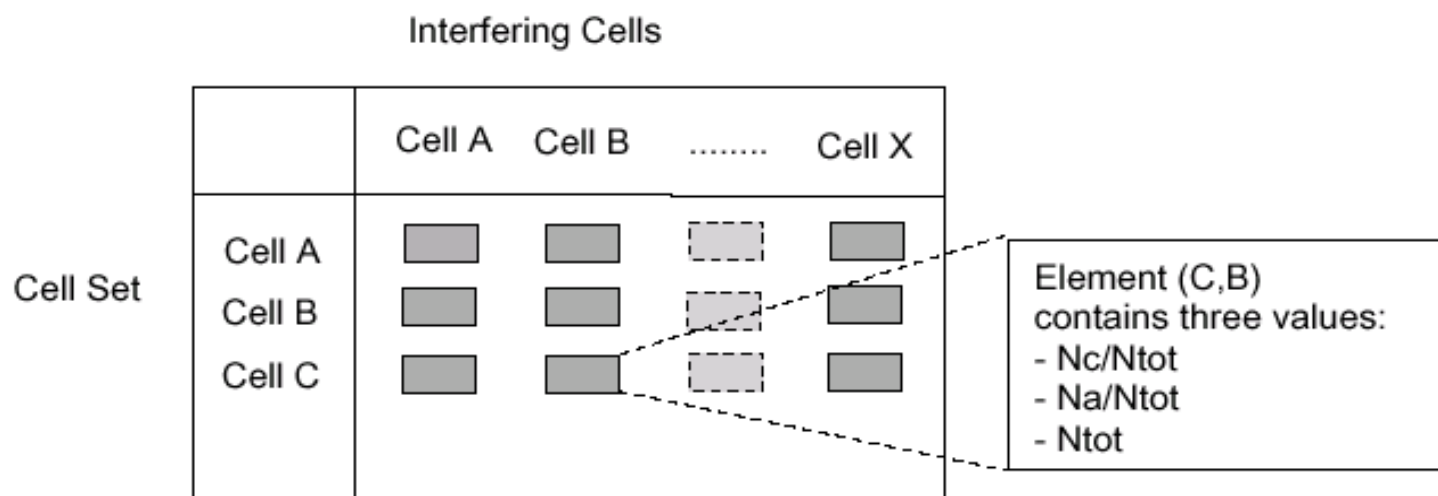
FAS downlink measurements

- based on measurements made by the mobile stations
- For every measurement report the signal strength of the reported cells are compared with the signal strength of the serving cell, to see if there is any potential interference between the cells.
- After a recording is made this information is compiled into a matrix that describes how well other cells are heard in a cell.
- The matrix is called the ICDM (Inter-Cell Dependency Matrix), and it has three values for each cell-relation; One value for co-channel frequencies, one for adjacent frequencies and one value for the number of measurement reports when the frequency was included in the active BA-list.
- The ICDM is independent of the frequency plan.

ICDM

- The result of the Downlink Recording is logically two Inter Cell Dependency Matrices (ICDMs). One represents co-channel interference and the other adjacent channel interference. The matrices are called ICDM-C and ICDM-A respectively. A matrix element in ICDM-C gives the estimated percentage interfered traffic in one cell caused by co-channel interference from another cell, if co-channel frequency is used in both cells. Analogously, a matrix element in ICDM-A gives the percentage interfered traffic in one cell caused by adjacent channel interference from another cell, if adjacent channel frequencies are used in both cells. To each matrix element a value is associated that represents the number of samples that the estimation is based on. In practise, since ICDM-C and ICDM-A are computed from the same measurement reports, they could be regarded as one ICDM with three values for each element: one giving the ICDM-C value, one giving the ICDM-A value and one giving the "number of counts" indication. The "number of counts" indicator is given by the total number of measurement reports received when the corresponding test frequency was included in the active BA list.

ICDM - Inter Cell Dependency Matrix

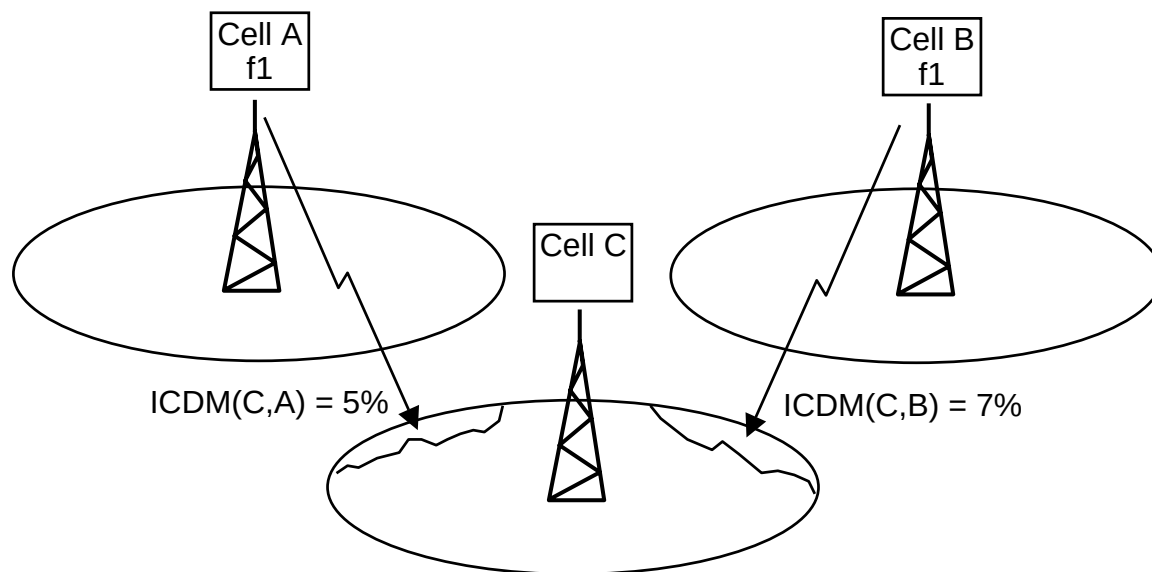


- Shows how much one specific cell interferes other cell
- ICDM is used to compute PIT values
- ICDM can be imported to TCP to simulate downlink interference instead of predicted data

PIT Percentage Interfered Traffic

- The ICDM is combined with the current frequency plan.
- For every frequency (both configured and non-configured) in every cell that was included in the recording, an estimation of the downlink interference for that frequency is calculated.
- This value is called PIT and shows how large percentage of the traffic that would be interfered if that frequency was used in the particular cell.

Percentage of Interfered Traffic



$$\text{PIT for } f1 \text{ in } C = 5 + 7 = 12\%$$

Percentage Interfered Traffic (PIT) is the downlink interference estimation for a specific frequency in a specific cell. It is estimated from the ICDM and the current frequency plan, where DTX, BTS Power Control and traffic load is taken into concern.

- Calculated for all frequencies in all cell

Creating a Downlink Recording

- A recording of the downlink interference can be ordered to create a new ICDM.
- When a new ICDM shall be created a number of options can be selected on whether it should be combined with an existing ICDM or not, and in the former case the way the ICDMs should be combined.

ICDM Settings alternatives

- Use an existing ICDM (when no downlink recording is performed).
- Use the ICDM that is produced during the recording (when Downlink recording is performed).
- Use a combination of a new ICDM (when Downlink recording is performed) and an existing in one of the following ways:
 - The new ICDM complements the existing ICDM, by completing missing information in the existing ICDM.
 - The existing ICDM complements the new ICDM where the new ICDM does not have any information.
 - The two ICDMs are added to each other, using a relative traffic weight.
- Specify whether the ICDM, that is measured during a downlink recording, is to be saved with a new name and a descriptive text or with the same name as the recording.

RNO – FAS ICDM Settings

ICDM

☐ **Use Existing ICDM**

ICDM Name	Relative SS Threshold for C/I (dB)	Relative SS Threshold for C/A (dB)	System Type	# Cells	Creation Date	Creator
FAS-01	-9.00	9.00	GSM900	40	2003-03-10 17:00	bjek
NISSE	-9.00	9.00	GSM900	0	2003-03-11 16:29	bjek
R3	-9.00	9.00		60	2002-03-27 10:18	jile
fasBsm?	-9.00	9.00	GSM900	40	2003-03-12 11:20	jile
foxAuto_2cells:1	-9.00	9.00	GSM900	40	2003-03-13 19:45	jile
foxr10	-9.00	9.00	GSM900	40	2003-03-10 15:15	jile
jilelcdm	-9.00	9.00	GSM900	770	2003-03-14 14:51	jile
largeFas	-9.00	9.00	GSM900	495	2003-03-17 10:40	bjek
nisse	-9.00	9.00	GSM900	40	2003-03-12 16:50	bjek

Combine ICDMs:
☒ **Weighted Accumulation**
☐ Use Recorded ICDM and use Selected ICDM to fill out Missing Elements
☐ Use Selected ICDM and use Recorded ICDM to fill out Missing Elements

Save ICDM
☒ **Use Recording Name as ICDM Name**
☐ **Name:**

BCCH Frequencies

All BCCH Frequencies Matching Cell Filter

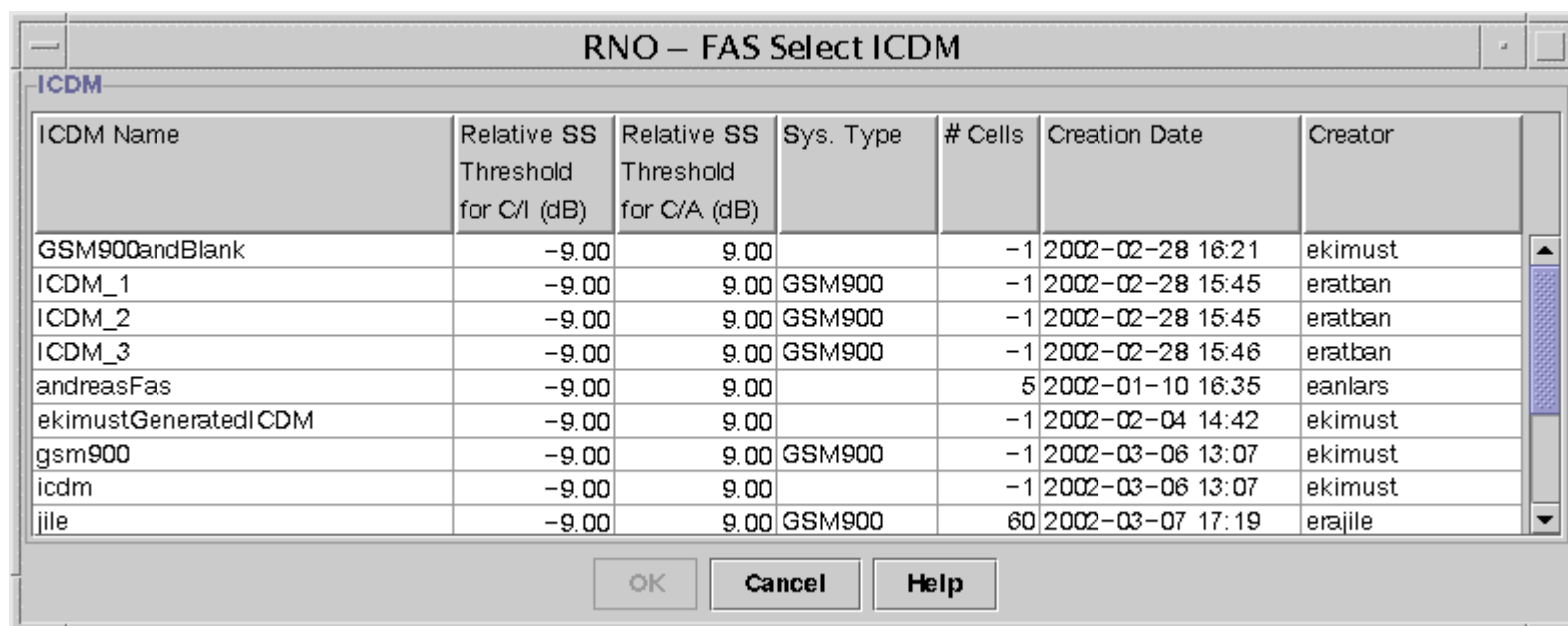
Multi Band Cell Settings

Other Band Factor:

At Measured ICDM Element 5%:
 At Measured ICDM Element 50%:

Comments

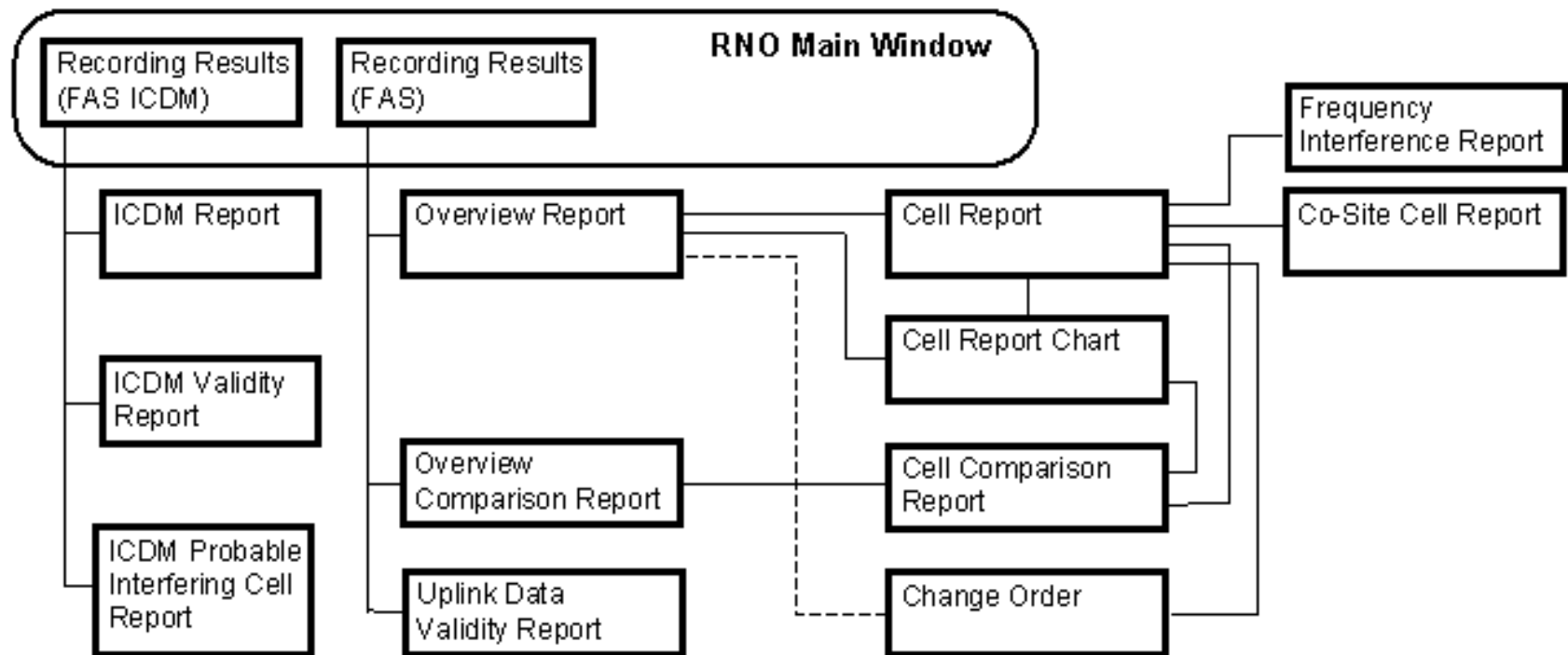
If an ICDM already exists and a PIT is to be estimated, the Select ICDM option makes it possible to select an existing ICDM from a dialogue box.



FAS - reports

- Overview Report
- Cell Report
- Cell Report Chart
- Overview Comparison Report
- Cell Comparison Report
- UL Data Validity Report

FAS report hierarchy

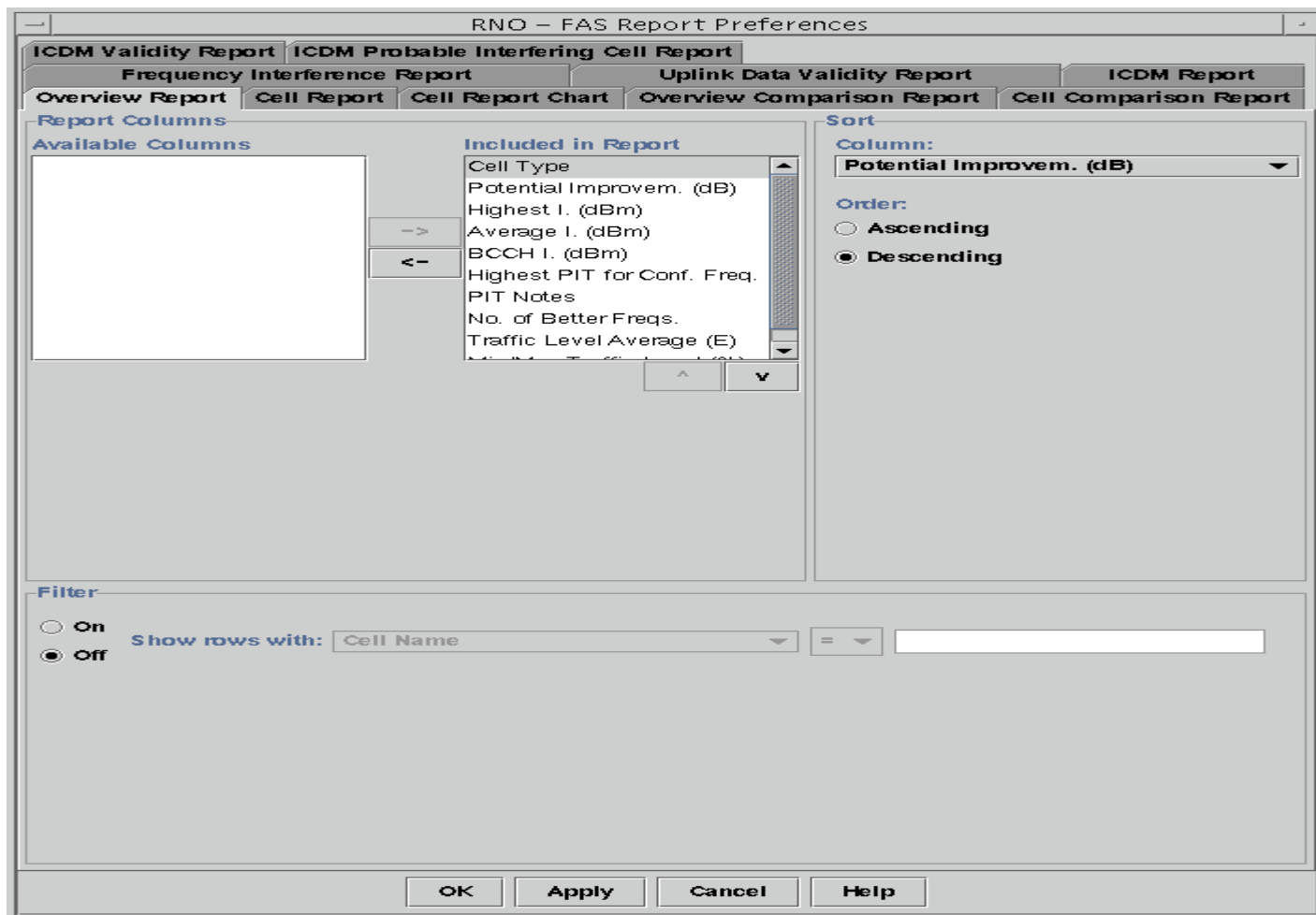


- **To run BAR or RIR-recordings manually and use RNO from the same OSS at the same time is not recommended and may cause inconsistencies**

The BSC limitations are the following:

- A FAS-recording is able to record on up to but not more than 150 frequencies.
- FAS is able to handle up to 10 active parallel FAS-recordings per BSC.
- **A cell is able to handle only one FAS recording period at a time.**

In the Report Preferences dialogue box you can limit the number of values included in the report.



RNO - FAS Overview Report										
File View Reports Tools										Help
Recording Name: fas_1227_2										
Date: 1999-12-27 -> 1999-12-27		Cell Set: Mobilia								
Days:		Frequencies: 41								
Hours: 12:10-12:20		Freq. Set: Mjolby900								
Percentile: 98.0		Creator: ossadm								
Cell Name	Cell Type	Potential Improvem. (dB)	Highest I. (dBm)	Average I. (dBm)	Perc. I. BCCH (dBm)	Max PIT for Conf. Freq.	PIT Notes	No. of Better Freqs.	Traffic Level Average (E)	Traffic Level Min/Max (%)
BMJOL01D/MJOL09R2		10.00	-91.50	-96.10	-98.50			12	13.64	100.00
BMJOL01D/MANT11R3		10.00	-94.50	-98.30	-97.50			15	17.48	100.00
BMJOL01D/MJOL23R1		10.00	-93.50	-97.00	-96.50			15	8.11	100.00
BMJOL01D/MJOL03R2		10.00	-92.50	-99.25	-98.50			7	46.95	100.00
BMJOL01D/MJOL13R3		9.00	-96.50	-102.25	-104.50			6	18.25	100.00
BMJOL01D/MJOL22R2		9.00	-93.50	-98.75	-100.50			9	11.54	100.00
BMJOL01D/MJOL09R1		8.00	-93.50	-96.33	-98.50			4	28.53	100.00
BMJOL01D/MANT11R2		8.00	-98.50	-100.25	-98.50			9	11.26	100.00
BMJOL01D/MJOL22R1		8.00	-95.50	-98.33	-97.50			8	23.51	100.00
BMJOL01D/MJOL21R1		8.00	-98.50	-100.50	-102.50			6	8.11	100.00
BMJOL01D/MJOL15R2		7.00	-93.50	-94.83	-94.50			7	7.18	100.00
BMJOL01D/MJOL23R2		7.00	-95.50	-99.64	-95.50			9	39.59	100.00
BMJOL01D/MJOL21R2		7.00	-95.50	-97.70	-99.50			6	15.42	100.00
BMJOL01D/MANT21R2		6.00	-104.50	-108.50	-110.50			17	12.63	100.00
BMJOL01D/MANT11R1		6.00	-91.50	-94.50	-93.50			9	16.58	100.00

RNO – FAS Cell Report

Help

File **View** **Reports** **Tools**

Cell Name: 5000/5000055
Recording Name: R3
Date: 2002-03-27 -> 2002-03-27
Days:
Hours: 10:10-10:30
Percentile: 95.0

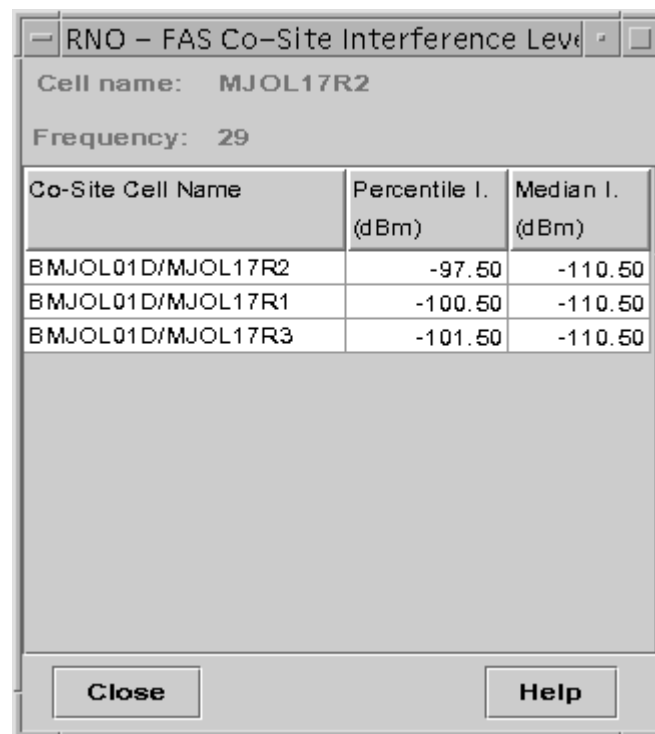
BCCH freq: 66
Cell Set: 5000_900_R3
Frequencies: 124
Freq. Set: Operators Full Set
Creator: jile
Traffic Level:

Configured Frequencies

Freq. No.	Cell Type	No. of Samples	Perc. I. (dBm)	Median I. (dBm)	PIT	PIT Notes	Warnings	Included in Freq. Set
78	normal	4187	-47.50	-78.50	0.16	Incomplete		
90	normal	34484	-80.50	-52.50	0.12	Incomplete		
66	normal	46978	-96.50	-102.50	3.85	Incomplete		

Test Frequencies

Freq. No.	No. of Samples	Perc. I. (dBm)	Median I. (dBm)	PIT	PIT Notes	Warnings	Included in Freq. Set
77	27758	-110.50	-84.50	9.23		A	
98	54810	-110.50	-72.50	36.80		E	
61	47847	-109.50	-72.50	0.00	Incomplete	E	
53	51152	-108.50	-66.50	24.24			
54	63675	-108.50	-80.50	4.54	Incomplete		
96	41293	-108.50	-96.50	0.03	Incomplete	C	
10	53053	-105.50	-93.50	102.00		C	



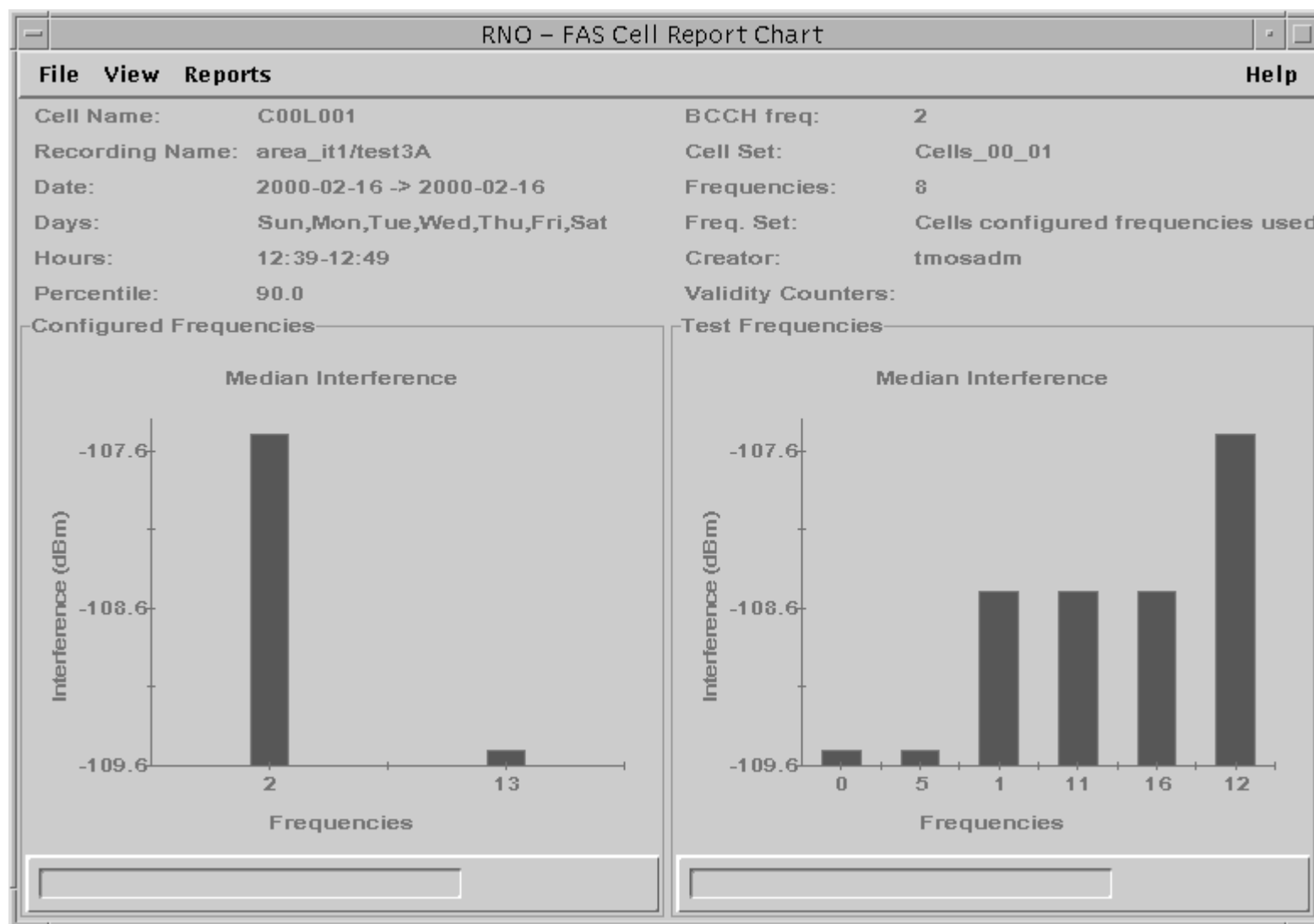
RNO - FAS Co-Site Interference Level

Cell name: MJOL17R2

Frequency: 29

Co-Site Cell Name	Percentile I. (dBm)	Median I. (dBm)
BMJOL01D/MJOL17R2	-97.50	-110.50
BMJOL01D/MJOL17R1	-100.50	-110.50
BMJOL01D/MJOL17R3	-101.50	-110.50

Close Help



RNO – FAS Uplink Data Validity Report		
File View		Help
Recording Name: R3		
Minimum Number of Uplink Samples: 500		
Number of Rows in Report: 55		
Creator: jile		
Cell Name	Frequency	No. of Uplink Samples
0000040	10	143
0000040	83	301
0000042	25	66
0000042	122	31
0000043	32	388
0000045	33	36
0000045	59	428
0000046	12	179
0000046	80	89
0000049	53	351
0000050	66	492
0000051	47	418
0000051	114	319
0000052	30	112
0000053	47	494
0000055	82	263
0000055	91	459
0000056	86	492
0000057	106	157
0000059	4	55
0000060	68	21

FAS - Cell Report

RNO - FAS Cell Report

File View Reports Tools Help

Cell Name: RYBSC1/RV1183

Recording Name: fas icdm 0418 even

Date: 2001-04-18 -> 2001-04-18

Days:

Hours: 16:30-19:30

Percentile: 95.0

BCCH freq: 72

Cell Set: 0_part0fLkpg_900

Frequencies: 78

Freq. Set: FOX 900 set

Creator: ericsson

Validity Counters: Traffic Level = 6.51 (E), Min/Max = 100.0 (%)

Configured Frequencies

Freq. No.	No. of Samples	Perc. I. (dBm)	Median I. (dBm)	PIT	PIT Notes	Warnings	Included in Freq. Set
91	6,498	-66.50	-89.50	8.12	Inc.	C,E,I	
57	6,498	-82.50	-102.50	0.82		C,E	
72	6,498	-82.50	-102.50	0.28	Inc.	E	
43	6,498	-85.50	-103.50	0.20	Inc.	C,E	
65	6,498	-89.50	-106.50	0.04		E	
36	6,498	-90.50	-106.50	0.36		E,I	
9	6,498	-92.50	-109.50	0.05		E	
95	6,498	-92.50	-108.50	0.00		E,I	
34	6,498	-94.50	-109.50	0.00		E,I	

View Warnings...

Test Frequencies

Freq. No.	No. of Samples	Perc. I. (dBm)	Median I. (dBm)	PIT	PIT Notes	Warnings	Included in Freq. S
24	6,498	-110.50	-110.50	0.90			
25	6,498	-110.50	-110.50	0.58			
26	6,498	-110.50	-110.50	0.73			
27	6,498	-110.50	-110.50	0.00			
28	6,498	-110.50	-110.50	0.00		E	
29	6,498	-110.50	-110.50	1.81		C	
19	6,498	-107.50	-110.50	48.14		C,E	
23	6,498	-106.50	-110.50	0.00			
30	6,498	-101.50	-110.50	1.00		E,I	
69	6,498	-100.50	-110.50	0.00		E,I,J	
20	6,498	-97.50	-110.50	3.18		E	
76	6,498	-90.50	-107.50	0.60		C,E,J	
53	6,498	-88.50	-102.50	0.00		C,E,I	
84	6,498	-88.50	-108.50	10.35		C,E	
94	6,498	-88.50	-107.50	8.84		A,C,E	

View Warnings...

04/ 038 13-LZU 108 5963

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GSM OSS Radio Network Optimizers

FAS - ICDM reports

- ICDM Report
- ICDM Validity Report
- ICDM Probable Interfering Cell Report

RNO – FAS ICDM Report

File Edit View Reports

Help

ICDM Name: R3

Creation Date: 2002-03-27

Relative SS Threshold for C/I: -9 dB

Relative SS Threshold for C/A: 9 dB

Creator: jile

Current BSC

5000

Current Cell

5000040

Potential Incoming Interference to Current Cell

Cell Type (current cell)	Cell Name	Co-Channel Interfered Traffic (%)	Adjacent Channel Interfered Traffic (%)	No. of Samples	Other Band Factor
normal	5001/5001059	48.28	6.74	10000	
normal	5001/5001051	48.26	3.15	10000	
normal	5001/5001043	44.05	1.37	10000	
normal	5001/5001049	43.38	4.00	10000	
normal	5001/5001079	40.97	6.50	10000	
normal	5001/5001061	38.27	1.74	10000	
normal	5001/5001069	37.33	6.27	10000	
normal	5001/5001041	37.15	2.51	10000	
normal	5001/5001099	36.37	2.08	10000	
normal	5001/5001073	35.34	5.20	10000	

Potential Outgoing Interference from Current Cell

Cell Name	Cell Type	Co-Channel Interfered Traffic (%)	Adjacent Channel Interfered Traffic (%)	No. of Samples	Other Band Factor
5000/5000045	normal	24.68	0.00	20000	
5000/5000049	normal	23.42	0.00	20000	
5000/5000056	normal	23.01	1.67	20000	
5000/5000054	normal	22.82	2.88	20000	
5000/5000062	normal	22.68	1.63	20000	
5000/5000076	normal	22.29	4.26	20000	
5000/5000066	normal	21.57	2.97	20000	
5000/5000073	normal	21.30	0.00	20000	
5000/5000085	normal	21.26	1.52	20000	
5000/5000070	normal	21.04	0.00	20000	

Comments

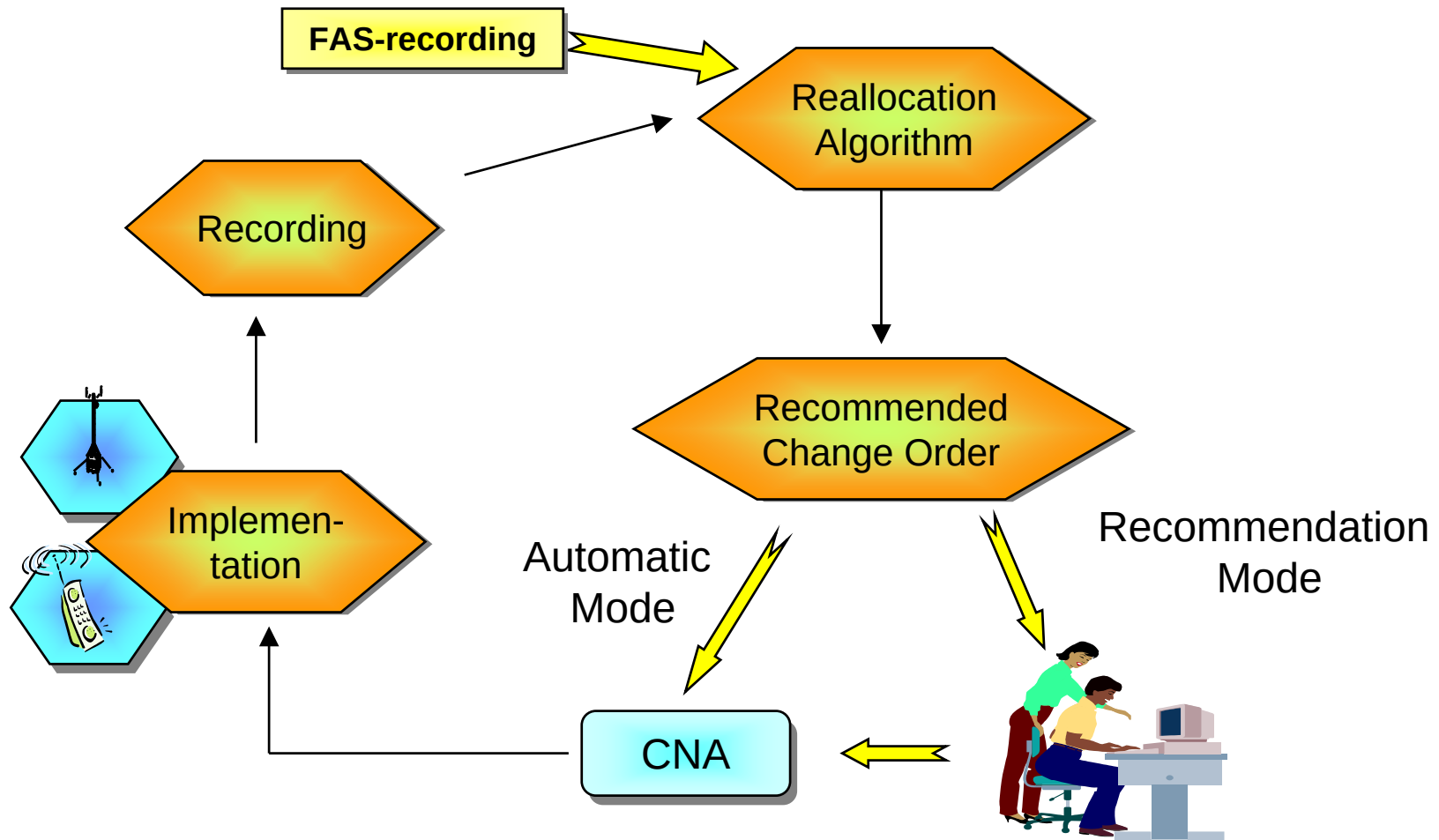
RNO – FAS ICDM Probable Interfering Cell Report								
File View Help								
ICDM Name: R3			Relative SS Threshold for C/I: -9 dB					
Creation Date: 2002-03-27			Relative SS Threshold for C/A: 9 dB					
			Creator: jile					
Current BSC			Current Cell					
5000 ▼			5000040 ▼					
BCCH	BSIC	Probable Cell	Predicted SS	Compensation (dB)	No. of Samples	Co-Channel Interfered Traffic (%)	Adjacent Channel Interfered Traffic (%)	Selected
4	31	5001/5001057	-44.97		10000	31.80	9.36 X	
5	30	5000/5000076	-74.68		20000	9.43	0.00 X	
6	31	5001/5001099	-75.78		10000	36.37	2.08 X	
10	31	5001/5001059	-40.18		10000	48.28	6.74 X	
11	30	5000/5000078	-75.41		20000	8.13	1.99 X	
16	31	5001/5001061	-40.17		10000	38.27	1.74 X	
17	30	5000/5000080	-76.10		20000	12.23	2.94 X	
21	30	5000/5000040	85.50		20000	4.59	3.41 X	
22	31	5001/5001063	-44.96		10000	30.25	1.84 X	
23	30	5000/5000082	-76.75		20000	18.98	2.24 X	
27	30	5000/5000042	-35.76		20000	12.51	1.12 X	
28	31	5001/5001065	-49.69		10000	24.08	4.46 X	
29	30	5000/5000084	-77.38		20000	4.58	4.84 X	
33	30	5000/5000044	-45.10		20000	21.06	4.13 X	
34	31	5001/5001067	-53.49		10000	4.30	0.00 X	
35	30	5000/5000086	-77.98		20000	15.63	4.03 X	
39	30	5000/5000046	-50.55		20000	10.85	0.00 X	
40	31	5001/5001069	-56.55		10000	37.33	6.27 X	
41	30	5000/5000088	-78.55		20000	16.41	2.21 X	
45	30	5000/5000048	-54.43		20000	3.27	2.22 X	
46	31	5001/5001071	-59.08		10000	32.50	2.64 X	
47	30	5000/5000088	-78.18		20000	18.58	1.54 X	
Data Is Filtered								

RNO - What does FOX do?

- Gives recommendation on suitable frequency reallocations (optimization)
- Can automatically update the network (implementation)
- Improve quality of the frequency plan
- Find suitable frequencies for new cells/TRXs
- Uplink interference statistics

- FOX always uses both uplink and downlink.
- Instead of just showing the results, FOX has the additional functions of proposing changes that will improve the network quality.
- This is done with an algorithm that takes all possible frequencies in all cells into account, and suggests the best changes for the cells included in the recording.
- There are two levels of FOX recordings:
 - Recommendation
 - FOX suggests a number of frequency reallocations that are based on uplink and downlink recordings. You can accept or reject the suggestions before they are implemented.
 - Automatic
 - all suggested changes is implemented without user interaction.

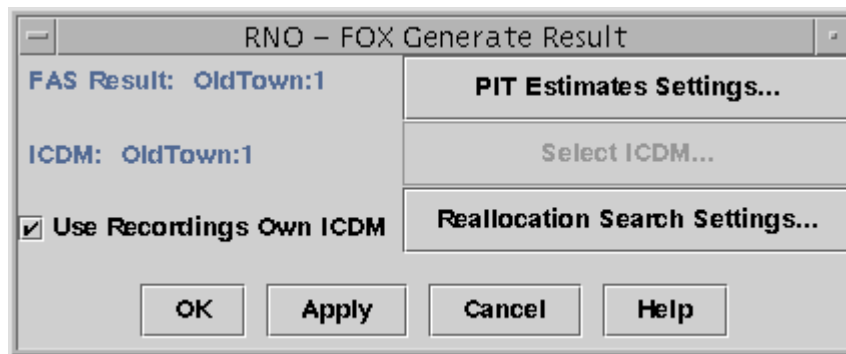
FOX - Frequency Optimization eXpert



Generating a FOX Result

It is possible to generate a FOX-result from an existing FAS-result. FOX then suggests changes to be made and you have to accept or reject the suggestions.

1. In the RNO main window select the Recording Results tab.
2. Select a FAS-result with the status Complete.
3. Click Tools -> FOX Generate Result.
4. Make your choices.
5. Click OK.



Generating a FOX Result- part2

- A new result is included in the list in the Recording Results tab.
- **The type is FOX Recom. and its status is Unconfirmed.**
- You can open a FOX Change Order Recommendation report and study the new recommendations for the selected result.

FOX - Reports

- Change Order Recommendation
- Change Order Evaluation
- Reallocation Log

FOX - Change Order Recommendation

RNO - FOX Change Order Recommendation											
File Edit View Reports Help											
Result Name: 0412 all 900:1,5											
Days: Percentile: 95.0											
Hours: 09:10-13:10 Reallocation Cell Set: 0_part0FLkpg_900											
Evaluation Start: 2001-04-12 03:10 Frequencies: 92											
Evaluation Stop: 2001-04-12 07:10 Freq. Set: Operators Full Set											
ICDM Name: fas icdm 0418 900 Creator: ericsson											
Change	Rank	Cell Name	Configured Frequency	Suggested Frequency	Suggested BSIC	Potential UL Improvement (dB)	Percentile UL I for Configured Freq. (dBm)	Percentile UL I for Suggested Freq. (dBm)	Potential DL PIT Improvement	DL PIT for Configured Freq.	DL PI* for Su Freq.
☒	1	VABSC1/VA1070	87	92		2.00	-108.50	-110.50	17.34	17.50	
☒	1	VABSC1/VA1410	68	66		0.00	-110.50	-110.50	9.81	10.89	
☒	1	VABSC1/VI1202	40	23		20.00	-90.50	-110.50	5.33	5.33	
☒	1	VABSC1/VI1202	54	28		19.00	-91.50	-110.50	5.35	5.35	
☒	1	VABSC1/VA1410	36	81		2.00	-108.50	-110.50	5.11	6.19	
☒	1	VABSC1/VI1333	37	88		3.00	-83.50	-86.50	3.28	3.40	
☒	1	RYBSC1/GO1420	57	79		0.00	-110.50	-110.50	3.06	3.49	
☒	1	RYBSC2/GO2173	37	89		5.00	-84.50	-89.50	2.90	3.64	
☒	1	RYBSC1/GO1330	87	76		0.00	-110.50	-110.50	2.53	2.97	
☒	1	VABSC1/VI1552	61	70		10.00	-90.50	-100.50	2.12	2.12	
☒	1	RYBSC1/GO1390	60	88		9.00	-96.50	-105.50	2.08	2.08	
☒	1	VABSC1/VI1160	85	76		3.00	-105.50	-108.50	2.25	2.25	
☒	1	VABSC1/VA1052	31	89		1.00	-82.50	-83.50	2.40	3.54	
☒	1	RYBSC2/GO2191	91	25		-2.00	-68.50	-66.50	2.29	2.29	
☒	1	VABSC1/VI1551	40	83		11.00	-90.50	-101.50	1.75	1.99	
☒	1	RYBSC2/GO2761	57	69		8.00	-82.50	-90.50	1.98	1.98	
☒	1	VABSC1/VA1051	73	83		5.00	-87.50	-92.50	1.59	1.93	
☒	1	RYBSC1/GO1181	63	69		13.00	-80.50	-93.50	0.68	0.68	
☒	1	VABSC1/VI1171	44	59		9.00	-85.50	-94.50	0.67	0.67	
☒	1	RYBSC1/GO1181	91	95		9.00	-81.50	-90.50	0.78	0.84	
☒	1	VABSC1/VA1012	90	20	56	25.00	-83.50	-108.50	0.00	0.00	
☒	1	VABSC1/VA1161	88	27	60	24.00	-85.50	-109.50	0.00	0.00	
☒	1	VABSC1/VI1022	43	23		23.00	-87.50	-110.50	0.19	0.19	
☒	1	RYBSC1/GO1113	64	23		23.00	-86.50	-109.50	0.15	0.15	

Data Is Filtered

Change Order Recommendations Report !!

The Edit menu is only included in the Change Order Recommendations Report for a result that has **status Unconfirmed** .

You have to **confirm or reject** the recommendation before a new recording period can be executed.

Change Order Evaluation !!!!!!!

- Select a **FOX-result with the status Confirmed** in the RNO main window, Recording Results tab and click on Reports -> FOX -> Change Order Evaluation to open the FOX Change Order Evaluation report.
- **A result with the status Confirmed means that the evaluation is ready, but the changes have not yet been updated in the network.**
- **A result with the status Complete means that the evaluation is ready and the results from the evaluation have been updated in the network.**
- The FOX Change Order Evaluation lists the cells which had their frequencies changed after implementation in the network. Those orders that were implemented are indicated in the Executed column.

RNO – FOX Change Order Recommendation

File Edit View Reports

Help

Result Name: jileFAS 3,1

Days:

Percentile: 95.0

Hours: 16:55-16:58

Reallocation Cell Set: 0701

Evaluation Start: 2002-03-07 16:55

Frequencies: 61

Evaluation Stop: 2002-03-07 16:58

Freq. Set: Operators Full Set

ICDM Name: jile

Creator: eardahl

Change	Rank	Cell Name	Configured Frequency	Suggested Frequency	Suggested BSIC	Potential UL Improvement (dB)	Percentile UL I for Configured Freq. (dBm)	Percentile UL I for Suggested Freq. (dBm)	Potential DL PIT Improvement	DL PIT for Configured Freq.	DL PIT for Suggested Freq.
☒	1	0701/0701098	47	61		40.00	-62.50	-102.50	4.62	4.69	
☒	1	0701/0701096	41	57		32.00	-71.50	-103.50	4.31	4.31	

Data Is Filtered

RNO – FOX Reallocation Log				
Recording Name: FoxOnePer_auto:1		Percentile: 90.0		
Date: 2000-05-11 -> 2000-05-11		Reallocation Cell Set: M00B00_15C_900		
Days:		Frequencies: 96		
Hours: 14:12-14:23		Freq. Set: Operators Full Set		
ICDM Name: FoxOnePer:1		Creator: svan		
Evaluation Start	Evaluation Stop	Applied to Network	No. of BCCH Reallocations	No. of TCH Reallocations
2000-05-11 14:12	2000-05-11 14:23		243 (251)	552 (568)

Close
Open Fox Change Order Evaluation...
Help

RNO – FAS Change Order

Frequencies to Delete:

Cell Name	Freq. No.	No. of Samples	Perc. I. (GSM)	Median I. (GSM)	PIT	PIT Notes	Warnings	Included in Freq. Set
5000/5000055	90	34484	-80.50	-52.50	0.12	Incomplete		

Frequencies to Add:

Freq. No.	No. of Samples	Perc. I. (GSM)	Median I. (GSM)	PIT	PIT Notes	Warnings	Included in Freq. Set	BSIC
77	27758	-110.50	-84.50	9.23		A		

Export to CNAI File...
Export to Planned Area...
Delete
Delete All
View Warnings...
Close
Help

FAS Recording Capacity

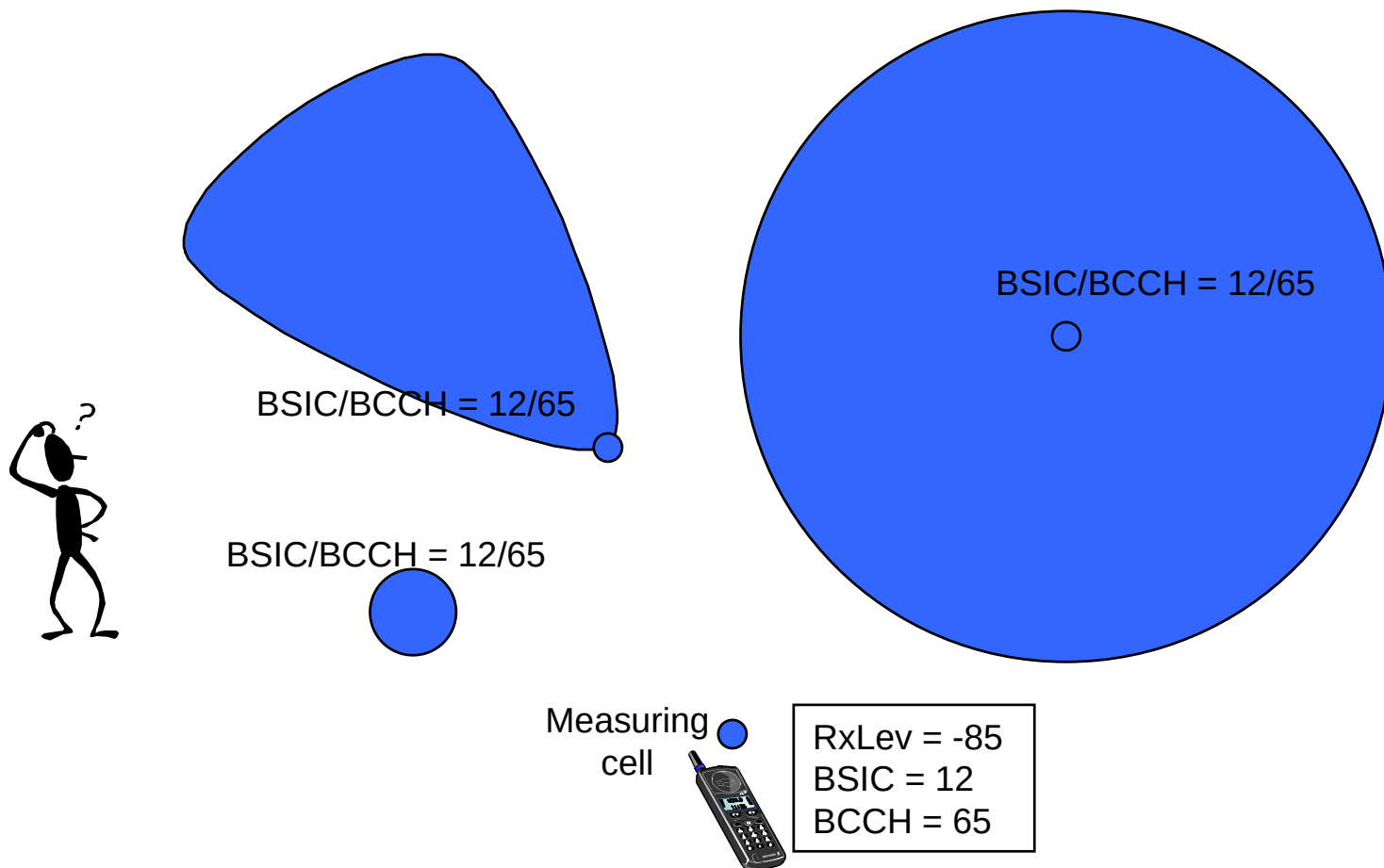
Resource	Capacity	Description
Number of simultaneous recordings under one OSS	10/BSC	Recordings are simultaneous if their recording Sessions overlap
Number of simultaneous recordings in one BSC	10	
Number of cells in one Cell Set	2000	
Number of frequencies in one recording	150	
Number of stored recording results per OSS	100	
Number of frequencies in a Frequency Set	150	
Number of Cell Sets that can be defined	100	
Number of Frequency Sets that can be defined	100	
Length of one recording	273 hours	After 273 hours overflow will occur for certain counters

FAS criteria

- **A FAS-recording period must not overlap (in time and cells) another FAS, FOX, NCS, NOX, or TET recording period**

- **The BA-List cannot be updated while a FAS downlink recording is going on.**

FAS/FOX - Finding the probable cell name



There are two types of matrices that are possible to create, view, modify, and print:

Barring Matrix

- The barring matrix indicates one cell and lists those cells to which the indicated cell cannot have the same frequencies allocated.

Spacing Matrix

- The spacing matrix defines the frequency spacing for frequencies allocated to the same cell, to co-sited cells and to neighbouring cells.
- The spacing can be indicated either as an integer or combiner spacing, which means that the frequencies allocated to the same cell must comply with the combiner spacing requirements.

FAS Spacing Matrix

RNO - FAS Spacing Matrix

Type Of Frequency/Cell	Own Cell	Co-Site Cell	Neighbour Cell
BCCH - BCCH		2 ▼	1 ▼
BCCH - TCH	combiner spacing ▼	1 ▼	1 ▼
TCH - TCH	combiner spacing ▼	1 ▼	0 ▼

OK Apply Print... Cancel Help